

MA378: Tutorial Sheet 1

with some solutions

These exercises are for tutorials. You do not have to submit solutions to these questions.

- Q1. (a) Is it always possible to find a polynomial of degree 1 that interpolates the single point (x_0, y_0) ? If so, how many such polynomials are there? Explain your answer.
- (b) Is it always possible to find a polynomial of degree 1 that interpolates the two points (x_0, y_0) and (x_1, y_1) ? If so, how many such polynomials are there? Explain your answer.
- (c) Is it ever possible to find a polynomial of degree 1 that interpolates the three points (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) ? If so, give an example.
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- Q2. For each of the following interpolation problems, determine (with explanation) if there is no solution, exactly one solution, or more than one solution. In all cases p_n denotes a polynomial of degree (at most) n . You are not required to determine p_n where it exists.

- (a) Find $p_1(x)$ that interpolates (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) , where $x_i = i - 1$ and $y_0 = 0$, $y_1 = -1$, $y_2 = 1$.

Answer: We want to interpolate the points $(-1, 0)$, $(0, -1)$ and $(1, 1)$ with a polynomial of degree 1. That is, we want to find a single straight line through these points. Since they are not co-linear, that is not possible: no solution exists.

- (b) Find $p_1(x)$ that interpolates (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) , where $x_i = i - 1$ and $y_0 = 0$, $y_1 = -1$, $y_2 = -2$.

Answer: We want to interpolate $(-1, 0)$, $(0, -1)$ and $(1, -2)$ with a polynomial of degree 1. These are co-linear: $y_i = -1 - x_i$. So exactly one solution exists.

- (c) Find $p_2(x)$ that interpolates (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) , where $x_i = i - 1$ and $y_0 = 0$, $y_1 = -1$, $y_2 = 1$.

Answer: Here $n = 2$ and we have three distinct points: $(-1, 0)$, $(0, -1)$ and $(1, 1)$. So standard theory applies (e.g., Theorems 2.3 and 2.7), so there exists exactly one solution.

- (d) Find $p_2(x)$ that interpolates (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) , where $x_i = (-1)^{i+1}$ and $y_0 = 0$, $y_1 = -1$, $y_2 = 1$.

Answer: Among the points we'd have to interpolate are $(x_0, y_0) = (-1, 0)$ and $(x_2, y_2) = (-1, 1)$. Since we can't have a polynomial with two different values at $x = -1$, there is **no** solution to this problem.

- (e) Find $p_2(x)$ that interpolates (x_0, y_0) and (x_1, y_1) where $x_i = (-1)^{i+1}$ and $y_0 = 0$, $y_1 = -1$.

Answer: Since there are only two points, $(x_0, y_0) = (-1, 0)$ and $(x_1, y_1) = (1, -1)$, and $n = 2$, there is an infinite number of solutions. For example, let p_1 be the usual degree 1 interpolant of these points. Then any quadratic of the form $p_1(x) + C(x + 1)(x - 1)$, for any C also interpolates them.

Q3. (From the 24/25 exam)

- (a) Let $f(x) = xe^x$. Calculate the Lagrange form of p_2 , the polynomial interpolant to f at the interpolation points $\{x_0, x_1, x_2\} = \{0, 1, 2\}$.

Answer: The Lagrange polynomials are

$$L_0(x) = \frac{1}{2}(x-1)(x-2), \quad L_1(x) = -x(x-2), \quad L_2(x) = \frac{1}{2}x(x-1).$$

Using the L_i , you can find p_2 as

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) = 0L_0(x) + eL_1(x) + 2e^2L_2(x),$$

which can be simplified as $p_2(x) = e(e-1)x^2 - e(e-2)x$.

- (b) Use Cauchy's Theorem to find an upper bound for $|f(x) - p_2(x)|$, at $x = 1.5$.

Answer: The error is given by

$$f(x) - p_2(x) = \frac{f'''(\tau)}{3!}(x-0)(x-1)(x-2),$$

for $\tau \in [0, 2]$. We calculate

$$f'''(x) = 3e^x + xe^x,$$

which is clearly an increasing function on $[0, 2]$, which is maximal at $x = 2$. Thus, $|f'''(\tau)| \leq |f'''(2)| = 5e^2$. Then

$$|f(x) - p_2(x)| \leq \frac{5e^2}{6}|x(x-1)(x-2)|,$$

so that

$$|f(1.5) - p_2(1.5)| \leq \frac{5e^2}{6}|(1.5)(0.5)(-0.5)| = \frac{5e^2}{16} \simeq 2.30908$$

Q4. Let's suppose there is something called a *clamped* polynomial interpolant, $\hat{p}_{n+2}(x)$, associated with a set of distinct points $\{x_0, x_1, \dots, x_n\}$, which has the properties that, for a given function f ,

- $\hat{p}_{n+2}$ has degree at most $n+1$.
- $\hat{p}_{n+2}(x_i) = f(x_i)$ for $i = 0, 1, \dots, n$.
- $\hat{p}'_{n+2}(x_0) = f'(x_0)$ and $\hat{p}'_{n+2}(x_n) = f'(x_n)$.

Show that, if such a polynomial exists, then it is unique.

Answer: Suppose there are two solutions, p and q . Set $s(x) = p(x) - q(x)$. So s is a polynomial of degree $n+2$. Since $s(x_i) = p(x_i) - q(x_i) = f(x_i) - f(x_i) = 0$, we see that s has as $n+1$ zeros: one at each node. Apply Rolle's theorem to see that $s'(x)$ is a polynomial of degree $n+1$, with n zeros in each of the intervals (x_0, x_1) , (x_1, x_2) , $\dots$, (x_{n-1}, x_n) . Note these are open intervals. We also have that $s'(x_0) = s'(x_{n+1}) = 0$. So, in fact, $s'(x)$ has $n+2$ zeros. Since it is a polynomial of degree $n+1$, it must be that $s'(x) \equiv 0$. That means $s(x)$ is constant, and since it has zeros, it must be zeros everywhere.

- Q5. Write down the linear spline l that interpolates $f(x) = \ln(x)$ at the points $x_0 = 1$, $x_1 = 1.5$ and $x_2 = 2$. Use this to estimate $\ln(x)$ at $x = 1.2$. How does this compare to the true value? Give an estimate for the maximum error, $\max_{1 \leq x \leq 2} |f(x) - l(x)|$, using Theorem 1.3 from Section 2.1 (Linear Splines).